

Effects of Digitalization on Labor Markets

- IT 250 – Digital Economy | Unit 5: Economics of Information

- Labor Market:

Represent the supply of labor (individuals seeking work) and demand for labor (employers hiring), interacting to determine wages, salaries, and employment levels

Introduction

- Digitalization changes how people work and find jobs
- Affects tasks, skills, and job types
- Technology can create new jobs but also replace old ones
- Example: Machines can do repetitive work, but humans now manage machines, data, and strategy

How Digitalization Changes Work

- Automates repetitive tasks
- Enables remote and flexible work
- Creates new job types
- Changes required skills
- Nepal Example: Freelance IT developers working remotely; online tutoring replacing classroom jobs
- Analogy: Like learning a new game — you need to understand the new rules to play

Job Displacement

- Some jobs disappear due to automation
- Routine office, clerical, and manual jobs affected most
- Nepal Example: Data entry jobs automated; travel agencies losing clients to online bookings
- Impact: Workers must reskill to stay employed

Job Creation

- New opportunities in AI, IT, digital marketing, e-commerce
- Remote work enables global freelancing
- Nepal Example: Developers on Upwork; YouTube content creators monetizing channels
- Analogy: Some doors close, new doors open — opportunities exist if you adapt

Skills Transformation

- Requires digital literacy, coding, data analysis, AI skills
- Soft skills: adaptability, communication, problem-solving
- Nepal Example: IT colleges teaching Python, AI, ML
- Analogy: Like learning the rules of a new game to play well

Labor Market Polarization

- High-skill jobs grow
- Low-skill routine jobs decline
- Middle-skill jobs shrink
- Nepal Scenario: Skilled programmers earn more, office clerks face uncertainty

Remote Work & Gig Economy

- Platforms enable remote and freelance work
- Flexible but less job security
- Nepal Example: Pathao drivers, Foodmandu delivery, Upwork freelancers
- Analogy: Being your own boss, but without guaranteed protection

Algorithmic Management

- Platforms monitor workers using AI, ratings, and task allocation
- Nepal Example: Foodmandu and Pathao track performance
- Impact: Productivity increases, autonomy decreases
- Analogy: Like a coach assigning positions — helps the team, limits personal choice

Effects on Wages

- Automation → routine jobs pay less
- High-skill digital jobs → pay more
- Nepal Example: Freelancers earning USD often earn more than local jobs
- Analogy: Some players score higher in new games, others fall behind

Job Quality & Work Conditions

- Gig economy offers flexibility but few benefits (health, retirement)
- Stress and uncertainty increase
- Nepal Example: Delivery workers face pressure to complete tasks quickly

Reskilling & Upskilling

- Workers must learn new skills to stay relevant
- Nepal Example: Government IT programs, private IT institutes
- Analogy: Like leveling up in a game — if you don't improve, you can't compete

Job Flexibility

- Remote work allows better work-life balance
- Nepal Example: Freelancers working from home while caring for family
- Positive: Increases accessibility to jobs

Job Insecurity

- Digital jobs can be unstable
- Platforms can terminate contracts anytime
- Nepal Example: Gig workers on ride-sharing apps depend on ratings
- Impact: Workers must constantly adapt and compete

Labor Market Transparency

- Platforms provide information about jobs, pay, and ratings
- Nepal Example: Freelancers compare Upwork vs Fiverr projects

Global Competition

- Workers compete globally, not just locally
- Nepal Example: Nepali developers competing with India, Pakistan, Philippines
- Analogy: Like an international sports tournament — you must perform better than everyone

Automation Risk

- Routine and predictable tasks at high risk
- Nepal Example: Accounting clerks, data entry workers
- Research Insight: OECD shows ~40% of routine jobs at risk globally

Algorithmic Bias & Fairness

- Platforms' AI can favor some workers over others
- Nepal Example: Delivery drivers rated lower for factors outside their control

Effects on Inequality

- Skilled workers benefit more
- Low-skill workers face job loss or wage stagnation
- Nepal Scenario: IT freelancers vs traditional clerks

Policy Implications

- Reskilling and training programs are crucial
- Social protection for displaced workers
- Digital labor laws for gig workers
- Nepal Example: IT training initiatives by government and private institutes

Positive Effects on Labor

- Access to global markets
- Flexible working hours
- Skill upgrading opportunities
- Nepal Example: Freelancers earning more than traditional local jobs

Negative Effects on Labor

- Job insecurity and wage gaps
- Pressure from algorithmic management
- Loss of low-skill jobs
- Nepal Example: Delivery app drivers facing performance pressure

Power Shift in Labor Markets

- Platforms and algorithms control task allocation
- Workers follow tasks and ratings → less autonomy
- Nepal Scenario: Pathao delivery tasks assigned automatically
- Key Question: Who really controls work — worker or platform?

Summary

- Machines replace some jobs, new jobs emerge
- Workers must adapt and reskill
- Platforms influence task allocation and visibility
- Digitalization reshapes labor market structure globally and in Nepal